Kaggle Competition

Objective

Review the Chemical Process Monitoring Time-Series dataset provided on Kaggle [Link].

Overview

This dataset is synthetic and multivariate. It simulates a continuous system of chemical reactors operating in two regimes: `A` and `B`. Sensor inputs, control dynamics, seasonal effects, and fault development are recorded as a time series. Samples were collected over approximately 180 days at 1-minute intervals. Within each regime, several reactors operate in parallel. About 6% of recorded values are missing to simulate sensor dropouts. Information from the Kaggle dataset is provided below:

Regime A: Baseline Operation

- Nominal Temperature/Pressure
- Efficient Cooling
- Lower Fault Probability
- Represents Stable Long-Term Operation

Regime B: Stress Operation

- Higher Temperature/Pressure
- Reduced Cooling Efficiency
- Increased Sensor Noise
- Higher Fault Probability
- Represents Harsh/High-Throughput Conditions

Multi-Reactor Review

For the reactors, the following rules apply:

- Same Process Logic
- Small Unit-Specific Differences

Data Dictionary

Category	Column Name	Description
Time & Identifiers	timestamp	Date and time when the sensor reading was recorded
	operating_regime	Operating condition (A = baseline, B = stressed)
	reactor_id	Unique identifier for each reactor unit
Environmental	ambient_temp_effect	Slowly varying simulated ambient temperature influence
Core Sensors	reactor_temp	Internal reactor temperature
	reactor_pressure	Internal reactor pressure
	feed_flow_rate	Material inflow rate to the reactor
	coolant_flow_rate	Cooling fluid flow rate
	agitator_speed_rpm	Mixing speed of the agitator
Process Performance	reaction_rate	Speed of the chemical reaction
	conversion_rate	Fraction of feed converted to product
	selectivity	Product purity or focus
	yield_pct	Overall process yield percentage
Mechanical & Electrical	vibration_rms	Root-mean-square vibration level
	motor_current	Electrical current drawn by the motor
	power_consumption_kw	Electrical power usage in kilowatts
Control Variables	temp_setpoint	Target temperature set by the control system
	pressure_setpoint	Target pressure set by the control system
Targets	fault_type	Fault class (0 = normal, 1–4 = fault types)
	efficiency_loss_pct	**Estimated percentage efficiency loss**
	time_to_fault_min	Minutes remaining before fault onset

Note: For this assessment, the `efficiency_loss_pct` target variable was evaluated to highlight loss during production.

Regime A was selected due to the reduced noise, increased cooling efficiency, and lower fault probability. EDA and feature engineering were conducted on Regime A's first reactor to serve as a blueprint for the others within the group.

Set Up

Import Libraries

In [1]:
```
# Import Libraries
import joblib
import matplotlib.pyplot as plt
import missingno as msno
import numpy as np
import pandas as pd
import seaborn as sns
import shap
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r
from sklearn.model_selection import train_test_split

# Set Seed
seed_1 = 42
```

Kaggle Dataset Loading

In [2]:
```
# Load Dataset
df_all = joblib.load("../data/dataset.joblib")
```

Exploratory Data Analysis (EDA)

Review Columns

In [3]:
```
# View Columns
df_all.columns
```

Out[3]:
```
Index(['timestamp', 'operating_regime', 'reactor_id', 'ambient_temp_e
ffect',
       'reactor_temp', 'reactor_pressure', 'feed_flow_rate',
       'coolant_flow_rate', 'agitator_speed_rpm', 'reaction_rate',
       'conversion_rate', 'selectivity', 'yield_pct', 'vibration_rm
s',
       'motor_current', 'power_consumption_kw', 'temp_setpoint',
       'pressure_setpoint', 'fault_type', 'efficiency_loss_pct',
       'time_to_fault_min'],
      dtype='object')
```

Note: There are redundancies within the dataset especially with the setpoint columns. For simplicity and clarity in this analysis, the reactor's performance will be evaluated without including the setpoints. However, for future work, both the setpoint values and their deviation from actual readings can be further examined to assess their potential impact on system performance.

Review Head

In [4]:
```
# Explore Head
df_all.head()
```

Out[4]:

	timestamp	operating_regime	reactor_id	ambient_temp_effect	reactor_tem
0	2024-01-01 00:00:00	A	A_R1	0.000000	181.13555
1	2024-01-01 00:01:00	A	A_R1	0.000485	182.24982
2	2024-01-01 00:02:00	A	A_R1	0.000970	183.12973
3	2024-01-01 00:03:00	A	A_R1	0.001454	183.02098
4	2024-01-01 00:04:00	A	A_R1	0.001939	183.07709

5 rows × 21 columns

Review Tail

In [5]:
```
# Explore Tail
df_all.tail()
```

Out[5]:

	timestamp	operating_regime	reactor_id	ambient_temp_effect	reacto
777595	2024-03-30 23:55:00	B	B_R3	NaN	196.
777596	2024-03-30 23:56:00	B	B_R3	-1.454452e-03	195.
777597	2024-03-30 23:57:00	B	B_R3	-9.696348e-04	196
777598	2024-03-30 23:58:00	B	B_R3	-4.848174e-04	194.
777599	2024-03-30 23:59:00	B	B_R3	-2.449294e-15	195.

5 rows × 21 columns

Note: The dataset is organized by time spanning two months ascending from 2024-01-01 00:00:00 to 2024-03-30 23:59:00. `timestamp` is formatted in ISO 8601 variation. For easier analysis and visualization, a `time_elapsed` column will be created to measure time passed since the start.

Review Reactor Information

In [6]:

```
# Get Names of Reactors
reactor_names = sorted(df_all["reactor_id"].unique().tolist())
reactor_names_A = [name for name in reactor_names if name.startswith('
operating_regime = sorted(df_all["operating_regime"].unique().tolist()

# See Reactor Information
print("Operating Regime:", operating_regime)
print("Reactor Names:", reactor_names)
print("Reactor Names, A:", reactor_names_A)
print("Reactor Count:", len(reactor_names))
```

```
Operating Regime: ['A', 'B']
Reactor Names: ['A_R1', 'A_R2', 'A_R3', 'B_R1', 'B_R2', 'B_R3']
Reactor Names, A: ['A_R1', 'A_R2', 'A_R3']
Reactor Count: 6
```

Note: `A_R1` is chosen as the template for EDA and feature engineering. Thus, the `operating_regime` column is not

necessary.

Select for Regime A Reactors

In [7]:
```
# Create List of Columns to Remove
col_sp = ['operating_regime', 'temp_setpoint', 'pressure_setpoint']

# Select Regime A's Values Only
df_A = df_all[df_all['operating_regime'] == 'A'].copy()

# Remove Columns
df_A = df_A.drop(columns=col_sp)

# Confirm Columns Removed
print([col not in df_A.columns for col in col_sp])
```

```
[True, True, True]
```

Review Statistics

In [8]:
```
# Show Statistics with 2 Decimals
df_A.describe().round(2)
```

Out[8]:

	ambient_temp_effect	reactor_temp	reactor_pressure	feed_flow_rate	c
count	365600.00	365285.00	365396.00	365354.00	
mean	-0.00	181.30	15.68	99.90	
std	7.07	1.14	0.11	1.52	
min	-10.00	177.79	15.21	77.48	
25%	-7.07	180.42	15.61	99.30	
50%	-0.00	181.11	15.68	99.99	
75%	7.07	182.22	15.75	100.67	
max	10.00	189.13	16.20	104.61	

Note: `vibration_rms` potentially suffers from large spikes. The mean is `1.51`, while the max is `3.62`, which indicates possible jumps in vibration or mechanical issues.

`time_to_fault_min` has a wide range, with a median of `469.00` and a standard deviation of `339.57`. This indicates high variance in time between faults. For `efficiency_loss_pct`, the mean is `0.54`% and the max is `25.00`%, which shows that rare efficiency losses can be severe.

Finally, for `reactor_pressure` and `reactor_temp`, the standard deviations are `7.07` and `1.14`, respectively, with `15.21` to `16.20` and `177.79` to `189.13` as the lower and upper bounds. This narrow range indicates stability.

Review Columns Size, Missing Info, and DataTypes

In [9]:
```
# Get Total Length
total_rows = len(df_A)

# Get DataType, Filled Data, Missing Count, and Missing Percentage
info = pd.DataFrame({
    'DType': df_A.dtypes.astype(str),
    'Non_Null': df_A.notna().sum(),
    'Missing_Count': df_A.isna().sum(),
    'Missing_Perc': (df_A.isna().sum() / total_rows * 100).round(2)
})

# Display Info
info['Missing_Perc'] = info['Missing_Perc'].astype(str) + '%'
display(info)
```

	DType	Non_Null	Missing_Count	Missing_Perc
timestamp	object	388800	0	0.0%
reactor_id	object	388800	0	0.0%
ambient_temp_effect	float64	365600	23200	5.97%
reactor_temp	float64	365285	23515	6.05%
reactor_pressure	float64	365396	23404	6.02%
feed_flow_rate	float64	365354	23446	6.03%
coolant_flow_rate	float64	365436	23364	6.01%
agitator_speed_rpm	float64	365316	23484	6.04%
reaction_rate	float64	365284	23516	6.05%
conversion_rate	float64	365656	23144	5.95%
selectivity	float64	365085	23715	6.1%
yield_pct	float64	365529	23271	5.99%
vibration_rms	float64	365677	23123	5.95%
motor_current	float64	365472	23328	6.0%
power_consumption_kw	float64	365626	23174	5.96%
fault_type	int64	388800	0	0.0%
efficiency_loss_pct	float64	365344	23456	6.03%
time_to_fault_min	float64	13223	375577	96.6%

Note: `NaN` values are present across several columns, with a particularly large amount in `time_to_fault_min`. It must be determined whether these values are random or clustered to decide how to address them.

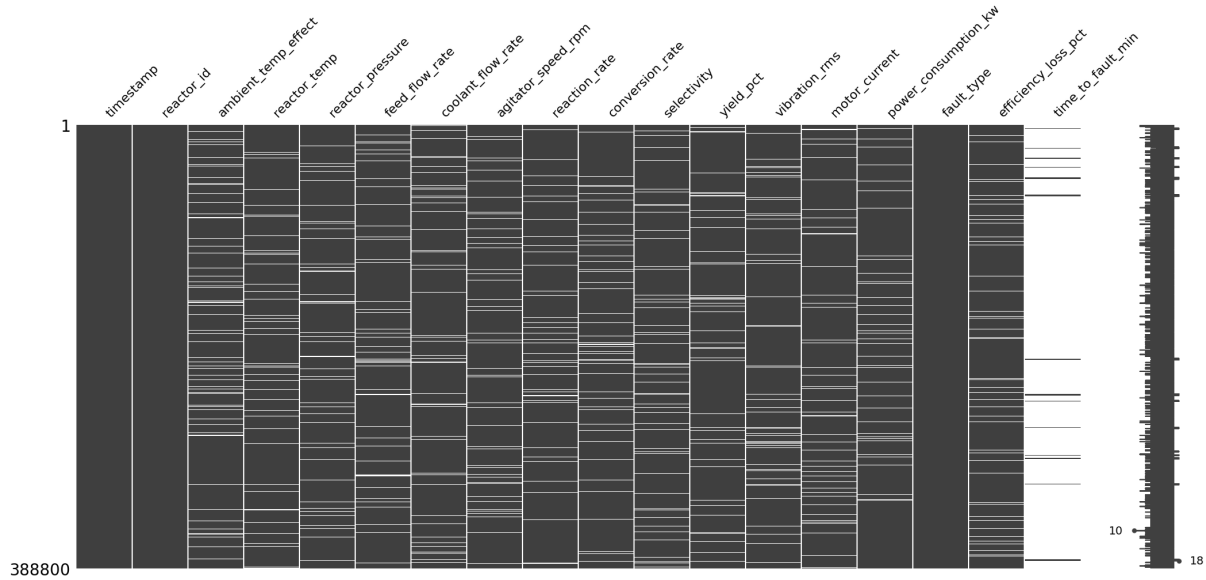
All `timestamp`, `reactor_id`, and `fault_type` values are accounted for. Most remaining columns have approximately `~6%` missing values due to simulated sensor noise. `time_to_fault_min` has the highest missingness at `96.62%`, indicating that these values are sparsely populated.

Data types include floats, integers, and objects. All data except the categorical variables and timestamps is `float64`. This simplifies operations within and across columns because no additional type conversion is required.

Visualize Missing Information

In [10]:
```
# Visualize the Missing Values
msno.matrix(df_A)
```

Out[10]: <Axes: >



Note: No clear correlation is observed between missingness in one column and missingness in others.

The sparseness of `time_to_fault_min` shows that values are distributed across time, with thicker bars indicating clustered recordings near fault events. This variable is not being evaluated here: only `3.3%` of data is available, and imputing it would create stepwise decay behavior that differs from the other continuous physical variables.

Although correlations could be computed, the data is too sparse and may mislead the model. Therefore, this column and the associated `fault_type` column will be removed. The remaining `~6%` missing values are sparsely distributed in sensor columns, supporting linear interpolation for imputation.

In [11]:
```
# Create List of Columns to Remove
col_fal = ['time_to_fault_min', 'fault_type']

# Remove 'operating_regime' Column (Unnecessary)
df_A = df_A.drop(columns=col_fal)

# Confirm Columns Removed
print([col not in df_A.columns for col in col_fal])
```

```
[True, True]
```

Helper Function for Plotting

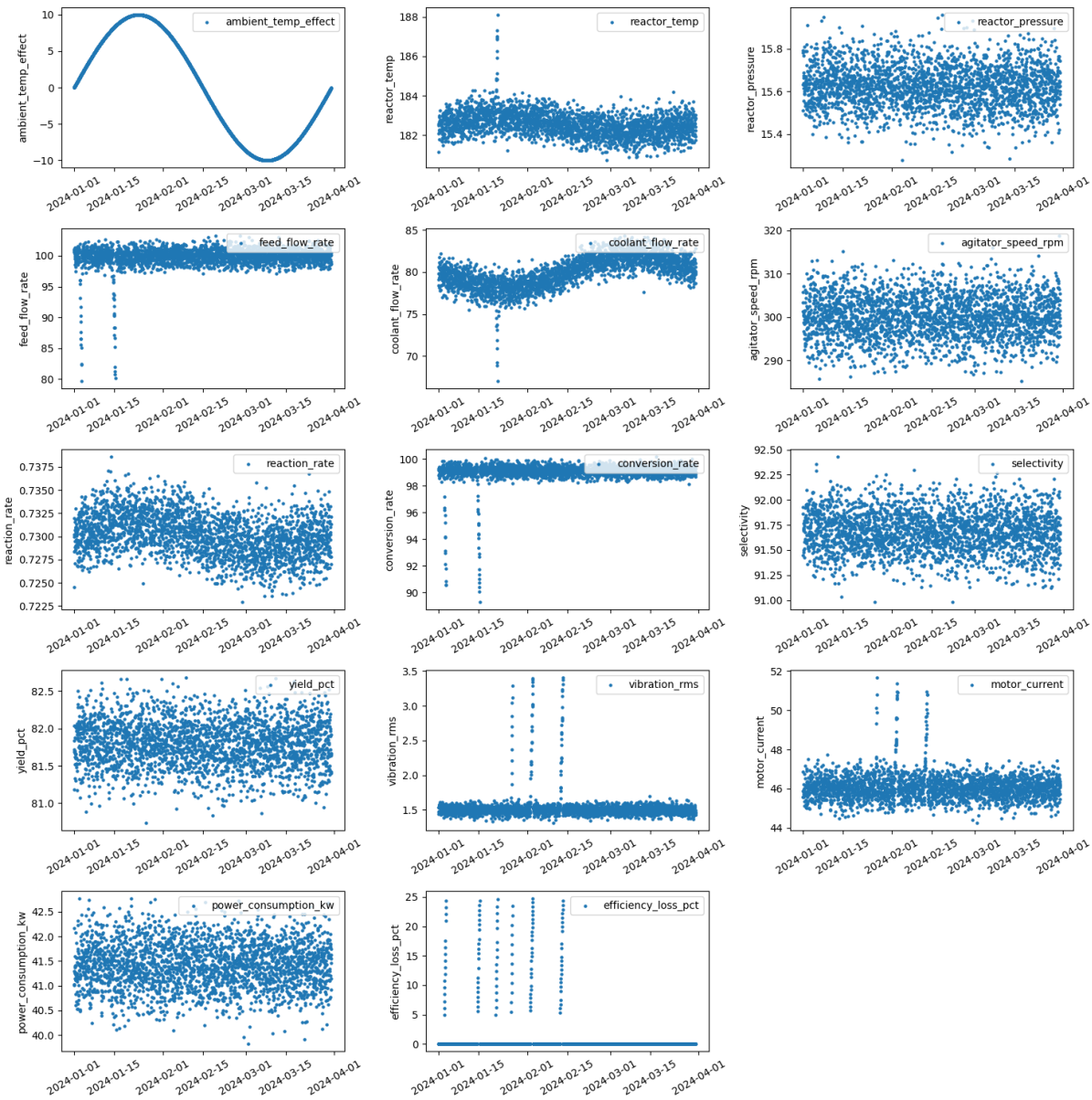
Below is a helper function for plotting the following graphs. It places all plots into a grid.

In [12]:
```
# Unified Grid Plotting Function
def unified_grid(data, columns, grid_type=2,
                 x_col=None, n_cols=3,
                 figsize=(5,3)):


    # Define the Rows, Axes
    n_rows = int(np.ceil(len(columns) / n_cols))
    fig, axes = plt.subplots(n_rows, n_cols, figsize=(figsize[0]*n_col
    axes = axes.flatten()    # For NumPy Axis Labeling

    # Loop Through and Depending on Argument Generate the Following Gr
    for row, col in enumerate(columns):

        # Create Scatter Plots
        if grid_type == 0:
            axes[row].scatter(data[x_col], data[col], label=col, s=5)
            axes[row].set_ylabel(col)
            axes[row].legend(loc='upper right')
            axes[row].tick_params(axis='x', rotation=30)

        # Create Box Plots
        elif grid_type == 1:
            axes[row].boxplot(data[col].dropna(),
                              vert=True)
            axes[row].set_title(col)
            axes[row].set_xticks([])
            axes[row].set_xticklabels([])

        # Create Histograms
        elif grid_type == 2:
            sns.histplot(data[col], kde=True, ax=axes[row])     # Ensu
            axes[row].set_title(f'Distribution of {col}')


    # Turn off Empty Grids
    for ax in axes[len(columns):]:
        ax.axis('off')

    plt.tight_layout()
    plt.show()
```

Scatter Plots

In [13]:
```
# Downsample for Plotting
```

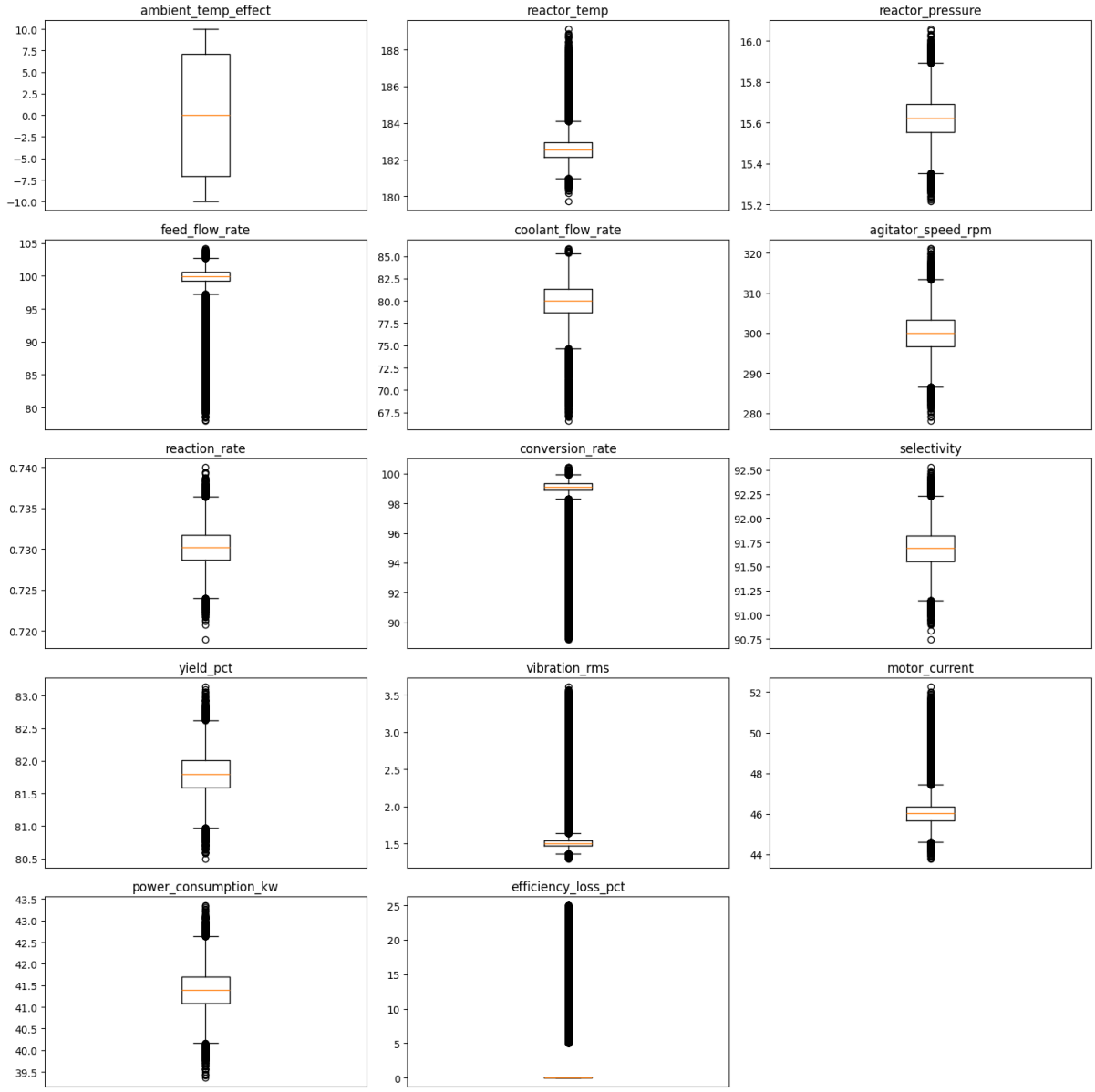
```
step = 50
reactor_id = 'A_R1'

# Define Time-Based and Categorical Column to Remove
plt_omit = ['timestamp', 'reactor_id']

# Remove Categorical and Timestamp Columns
params = [col for col in df_A.columns
          if col not in
          plt_omit]


df_reactor = df_A[df_A['reactor_id'] == reactor_id].copy()
df_downsampled = df_reactor.iloc[::step].copy()

# Convert timestamp to datetime for better formatting
if 'timestamp' in df_downsampled.columns:
    df_downsampled.loc[:, 'timestamp'] = pd.to_datetime(df_downsampled

# Use unified_grid for scatter plots
unified_grid(df_downsampled, params, grid_type=0, x_col='timestamp', n
```



Note: `ambient_temp_effect` displays a sinusoidal pattern as does `reaction_rate`, `reactor_temp`, and `coolant_flow_rate`. This is consistent with the expectation that cyclical changes in ambient temperature would drive corresponding fluctuations. The other sensors remain relatively stable and show plateau-like behavior with some stochastic variation. Occasional abrupt jumps may indicate potential sudden failures.

Boxplots

In [14]:
```
# Plot
unified_grid(df_reactor, params, grid_type=1, n_cols=3, figsize=(5,3))
```

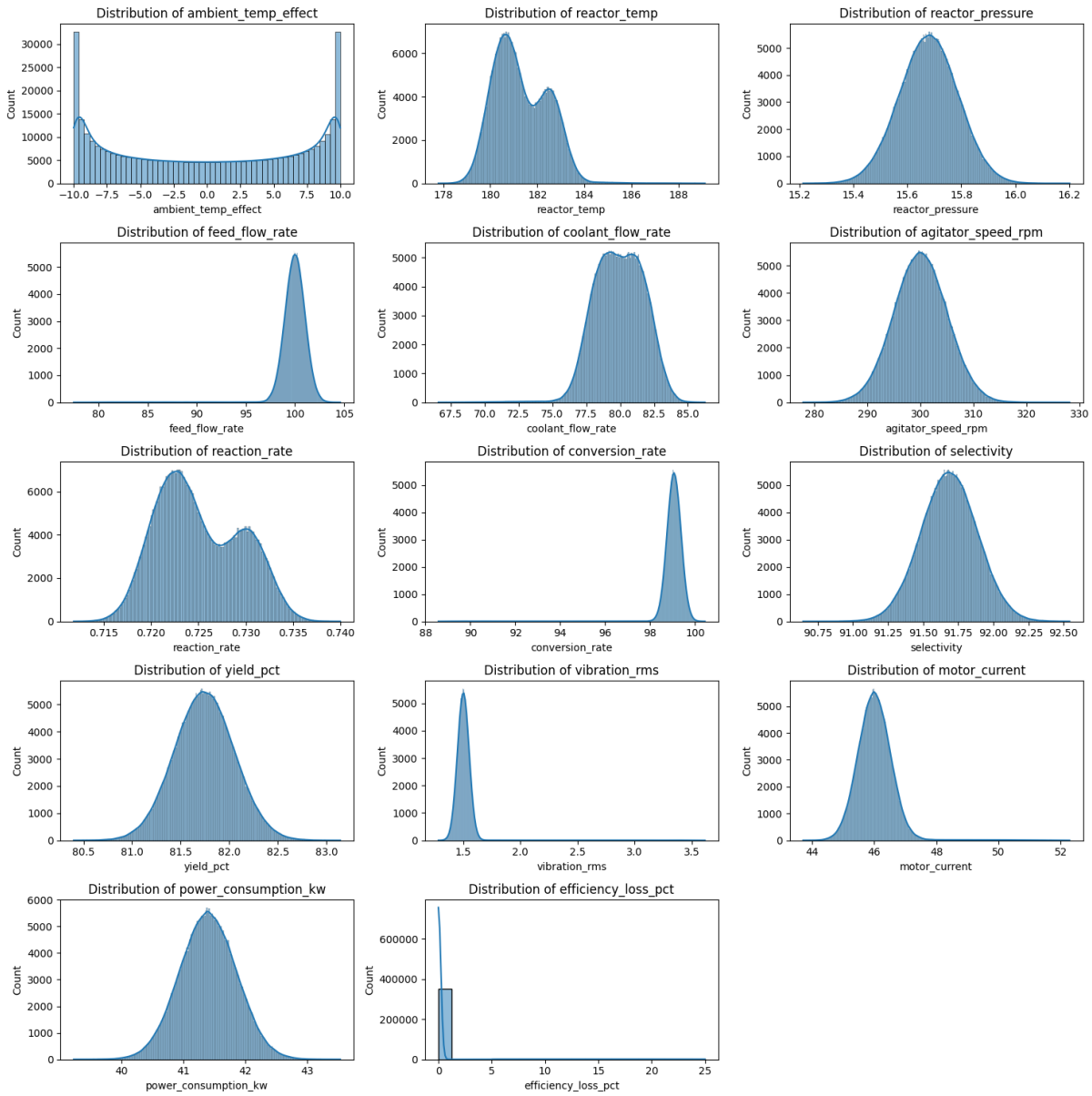


Note: `ambient_temp_effect` again mirrors a predictable pattern shown in the scatter plot with a reliably distributed boxplot. There are few outliers, which implies that this parameter, although oscillating, is predictably doing so, which points to machine health in this regard.

The rest of the sensors have thin boxplots with several outliers that surpass the minimum and maximum values frequently. For `reactor_temp`, `efficiency_loss_pct`, `motor_current`, and `vibration_rms`, the outliers are skewed above the maximums, whereas for `feed_flow_rate`, `coolant_flow_rate`, and `conversion_rate`, the opposite trend is observed. This potentially indicates positive or negative correlations that will be investigated with a heatmap.

Sample of Distribution Plots

In [15]:
```
# EDA: Distribution plots for numeric columns in a grid
unified_grid(df_A, params, grid_type=2, n_cols=3, figsize=(5,3))
```

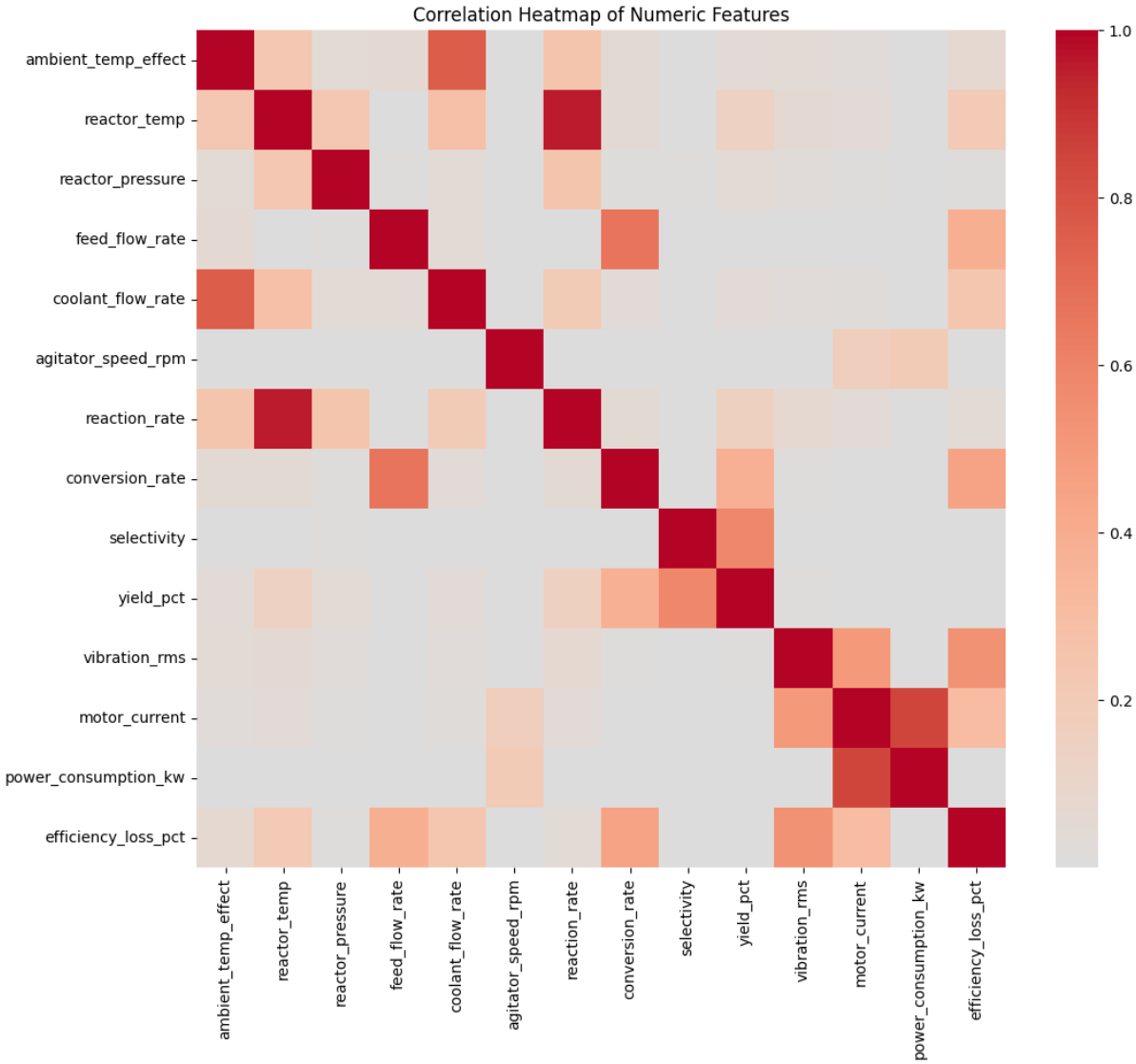


Note: Reaction rate, temperature, and flow rate variables show multimodal distributions which may indicate nonlinear relationships or multiple operating regimes. Tree-based models can capture nonlinear interactions and are robust to outliers (as observed in the boxplots) and multimodal feature distributions. Thus, a tree-based modeling approach will be used.

Heatmap

In [16]:
```
# Get Numerical Columns
numeric_cols = [col for col in df_A.columns if col not in plt_omit]

# Create Correlation Matrix
corr = df_A[numeric_cols].corr().abs()
```

```
upper_triangle = corr.where(np.triu(np.ones(corr.shape), k=1).astype(b

# Plot Heatmap
plt.figure(figsize=(12,10))
sns.heatmap(corr, cmap='coolwarm', center=0)
plt.title('Correlation Heatmap of Numeric Features')
plt.show()
```



Note: The heatmap shows strong correlations between the columns that are consistent with physical principles. Some include `reactor_temp` and `reaction_rate` (as expected through the `Arrhenius Equation`), `vibration_rms` and `efficiency_loss_pct` (`energy conservation`), `motor_current` and `power_consumption_kw` (as expected through `Ohm's Law`), as well as `conversion_rate` with `feed_flow_rate` (`mass balance`). The correlation matrix will be further evaluated numerically. Features will be removed if the correlation is above `0.95` to reduce redundancy.

Preprocessing

Interpolate Missing Values

In [17]:
```
# Apply Interpolation

# Remove Unneeded Columns for Interpolation
int_omit = ['timestamp', 'reactor_id']
cols_omit = [col for col in df_A.columns if col not in int_omit]

# Apply Interpolation
for col in cols_omit:
    df_A[col] = df_A[col].interpolate(method='linear')
```

Note: Since this dataset comes from a physical and continuous system, missing values can be imputed using linear interpolation along the time axis to preserve the natural trend.

Interpolation Confirmation

In [18]:
```
# Confirm Columns Have Been Filled
missing_summary = df_A.isna().sum()

# Show
print("Missing Values:")
print(missing_summary)
```

```
Missing Values:
timestamp               0
reactor_id              0
ambient_temp_effect     0
reactor_temp            0
reactor_pressure        0
feed_flow_rate          0
coolant_flow_rate       0
agitator_speed_rpm      0
reaction_rate           0
conversion_rate         0
selectivity             0
yield_pct               0
vibration_rms           0
motor_current           0
power_consumption_kw    0
efficiency_loss_pct     0
dtype: int64
```

Feature Engineering

Time-based Features (Elapsed Time)

Create a `time_elapsed` column from `timestamp`. This represents elapsed time since the start in a numeric format that the tree model can ingest. The `timestamp` column will be removed afterward because it is no longer needed.

In [19]:
```
# Convert Timestamp to Time Elapsed Column
df_A['timestamp'] = pd.to_datetime(df_A['timestamp'])
start_time = df_A['timestamp'].min()
df_A['time_elapsed'] = ((df_A['timestamp'] - start_time).dt.total_seco

# Sort by Time Elapsed
df_A = df_A.sort_values('time_elapsed').reset_index(drop=True)

# Remove Timestamp Column
df_A = df_A.drop(columns=["timestamp"])
```

In [20]:
```
# Confirm Time Elapsed Column
df_A['time_elapsed'].head(6)
```

Out[20]:
```
0     0.0
1     0.0
2     0.0
3    60.0
4    60.0
5    60.0
Name: time_elapsed, dtype: float64
```

Selecting for Lag and Window

This is a time-series model, so lag features and rolling statistics are needed for the tree to capture temporal relationships. Here, the amount of lag is evaluated to determine whether adding more lag features yields meaningful improvement or diminishing returns.

In [21]:
```
# Create df for Lag based on Regime A's df
df_lag = df_A[df_A['reactor_id'] == 'A_R1'].copy()

# Set Up Foundation for New Lag Table
col_corr = ['efficiency_loss_pct']
corr_matrix = pd.DataFrame()

# Check Lag Features
for lag in range(1, 10):

    # Shift and Obtain Result for Lag Column
    col_name = f'lag_{lag}'
```

```
    df_lag[col_name] = df_lag['efficiency_loss_pct'].shift(lag)

    # Calculate Correlation for Lag
    corr_value = df_lag['efficiency_loss_pct'].corr(df_lag[col_name])

    # Append to Lag Column Table
    col_corr.append(col_name)

# Get Correlation Matrix for Lags
corr_matrix = df_lag[col_corr].corr()
print(corr_matrix['efficiency_loss_pct'])
```

```
efficiency_loss_pct    1.000000
lag_1                  0.998760
lag_2                  0.997512
lag_3                  0.996263
lag_4                  0.995015
lag_5                  0.993767
lag_6                  0.992519
lag_7                  0.991271
lag_8                  0.990023
lag_9                  0.988774
Name: efficiency_loss_pct, dtype: float64
```

Note: Lower lag numbers are more highly correlated with the target. There are diminishing returns after lag 4. Adding more lag columns increases computation cost and can introduce noise or overfitting. More than one lag variable is needed for rolling standard deviation, so lags 1-3 are selected.

Lag Features and Rolling Statistics

In [22]:
```
# Define Lags and Window Size for Rolling Statistics
useful_lags = [1, 2, 3]
window_size = 3

# Create New df for Stats and Dictionary for New Columns
df_stat = df_A.copy()
col_stat = {}

# For Each Column, Except non-float Dtypes, Get Rolling Statistics and
for col in df_A.columns:
    if col not in ['timestamp', 'reactor_id']:

        #  Calculate Rolling Statistics, min_periods = 1 Ensures Mean
        col_stat[f'{col}_roll_mean'] = df_A[col].rolling(window=window
        col_stat[f'{col}_roll_std']  = df_A[col].rolling(window=window

        #  Get Lag Features
        for lag in useful_lags:
            col_stat[f'{col}_lag{lag}'] = df_A[col].shift(lag)
```

```
# Convert to DataFrame
df_stat_new = pd.DataFrame(col_stat)
df_stat = pd.concat([df_stat, df_stat_new], axis=1)
```

Interaction features

This step evaluates whether combining relationships between features can improve model performance.

In [23]:
```
# Set Standard Correlation Thresholds
thres_low = 0.3
thres_high = 0.95

# Get Numerical Columns
numeric_cols = [col for col in df_A.columns if col not in plt_omit]

# Create Correlation Matrix
corr = df_A[numeric_cols].corr().abs()
upper_triangle = corr.where(np.triu(np.ones(corr.shape), k=1).astype(b

# Set Up Dictionary for Key:Value Pairs
corr_pairs = {}

# Iterate Through Upper Triangle, Ignore Target Column
for col1 in [col for col in upper_triangle.columns if col != 'efficien
    for col2 in [col for col in upper_triangle.index if col != 'effici

        # Sort by Column Names to Ensure Uniqueness
        key = tuple(sorted([col1, col2]))

        # If Not Evaluated Yet
        if key not in corr_pairs:

            # Assign Value
            val = upper_triangle.loc[col2, col1]

            # If Meets Threshold, Evaluate
            if thres_low <= val <= thres_high:
                corr_pairs[key] = val

# Sort Correlation Pairs
sorted_data = dict(sorted(corr_pairs.items(), key=lambda item: item[1]

# Save into DataFrame
df_corr = pd.DataFrame(sorted_data.items(), columns=['Feature Pair', '
df_corr = df_corr.sort_values(by='Correlation', ascending=False).head(

# Check the correlation result
print(df_corr)
```

```
                                Feature Pair  Correlation
8             (reaction_rate, reactor_temp)     0.943466
7       (ambient_temp_effect, time_elapsed)     0.779688
6  (ambient_temp_effect, coolant_flow_rate)     0.769116
5     (motor_current, power_consumption_kw)     0.768592
4         (conversion_rate, feed_flow_rate)     0.672789
3         (coolant_flow_rate, time_elapsed)     0.589563
2                  (selectivity, yield_pct)     0.527716
1            (motor_current, vibration_rms)     0.505118
0              (conversion_rate, yield_pct)     0.343819
```

Note: Investigating whether to remove one feature from the pair `('reaction_rate', 'reactor_temp')`, since the correlation value (`0.94`) is close to the `0.95` threshold, as expected from the `Arrhenius Equation`.

`('motor_current', 'power_consumption_kw')` are physically related and therefore redundant as separate feature columns, as expected from `Ohm's Law`.

Remove Power-Current Feature Column

In [24]:
```
# Delete Unneeded Feature Column
del corr_pairs[tuple(sorted(['motor_current', 'power_consumption_kw']))]
```

Explore Reactor-Target Correlations

In [25]:
```
# Review 1st Column's Impact on Target
value = upper_triangle.loc["reaction_rate", "efficiency_loss_pct"]
print("Reaction Rate x Efficiency Loss:", value)

# Review 2nd Column's Impact on Target
value = upper_triangle.loc["reactor_temp", "efficiency_loss_pct"]
print("Reactor Temp x Efficiency Loss: ", value)
```

```
Reaction Rate x Efficiency Loss: 0.04691097492962186
Reactor Temp x Efficiency Loss:  0.20977485609639637
```

Note: The `reaction_rate` column has less influence on `efficiency_loss_pct`. Therefore, it will be removed instead of `reactor_temp`. The interaction feature for `('reaction_rate', 'reactor_temp')` will also be removed due to its high correlation, which would introduce redundancy in the tree model.

In [26]:
```
# Remove Reaction Rate (Feature) Column
del corr_pairs[tuple(sorted(['reaction_rate', 'reactor_temp']))]
```

```
df_A = df_A.drop(columns=["reaction_rate"])

# Remove from Statistics DataFrame
cols_rmv = [col for col in df_stat.columns if 'reaction_rate' in col]
df_stat = df_stat.drop(columns = cols_rmv)
```

Save to DataFrame

In [27]:
```
# Prepare DataFrame
df_fe = df_A.copy()

# Compute the Feature Columns for Appropriate Correlation Pairs
for (col1, col2), corr_val in corr_pairs.items():

    # Create Interaction Feature
    df_fe[f"{col1}_x_{col2}"] = df_A[col1] * df_A[col2]

# View Feature Engineering DataFrame
df_fe.head()
```

Out[27]:

	reactor_id	ambient_temp_effect	reactor_temp	reactor_pressure	feed_flow_
0	A_R1	0.000000	181.135558	15.791013	101.10
1	A_R3	0.000242	179.850553	15.884923	99.60
2	A_R2	0.000000	181.174608	15.747025	101.51
3	A_R1	0.000485	182.249820	15.706975	98.93
4	A_R3	0.000485	180.236922	15.738926	100.30

5 rows × 22 columns

Combine Feature Engineering, Stats, and Regime A's DataFrame

In [28]:
```
# Assure All are Descending By reactor_id and time_elapsed
sort_cols = ["time_elapsed", "reactor_id"]
df_A = df_A.sort_values(sort_cols, ascending=[True, True]).reset_index
df_stat = df_stat.sort_values(sort_cols, ascending=[True, True]).reset
df_fe = df_fe.sort_values(sort_cols, ascending=[True, True]).reset_ind

# Combine All Columns Together
df_combined = df_A.copy()
for df in [df_stat, df_fe]:
    for col in df.columns:
        df_combined[col] = df[col]

# Drop Rows with Lag or Rolling Stats NaN Values
```

```
df_combined = df_combined.dropna().reset_index(drop=True)
```

Confirm that Column Count Matches

In [29]:
```
# Print Column Sizes
names = ["df_A", "df_stat", "df_fe", "df_combined"]
lengths = [len(df_A.columns), len(df_stat.columns), len(df_fe.columns)

pd.DataFrame(zip(names, lengths), columns=["Column", "Lengths"])
```

Out[29]:

	Column	Lengths
0	df_A	15
1	df_stat	85
2	df_fe	22
3	df_combined	92

Note: The original DataFrame has `15` columns. The Lag DataFrame adds `70` unique columns (`85` – `15`), and the feature engineering DataFrame adds `7` unique columns (`22` – `15`). Combined, this gives a total of `92` columns, which matches the sum of `15` , `70` , and `7` .

Review New DataFrame

In [30]:
```
# Review
df_combined.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 388797 entries, 0 to 388796
Data columns (total 92 columns):
 #   Column                                      Non-Null Count   Dtype
---  ------                                      --------------   -----
 0   reactor_id                                  388797 non-null  object
 1   ambient_temp_effect                         388797 non-null  float64
 2   reactor_temp                                388797 non-null  float64
 3   reactor_pressure                            388797 non-null  float64
 4   feed_flow_rate                              388797 non-null  float64
 5   coolant_flow_rate                           388797 non-null  float64
 6   agitator_speed_rpm                          388797 non-null  float64
 7   conversion_rate                             388797 non-null  float64
 8   selectivity                                 388797 non-null  float64
 9   yield_pct                                   388797 non-null  float64
 10  vibration_rms                               388797 non-null  float64
 11  motor_current                               388797 non-null  float64
 12  power_consumption_kw                        388797 non-null  float64
```

```
 13  efficiency_loss_pct                      388797 non-null  float64
 14  time_elapsed                             388797 non-null  float64
 15  ambient_temp_effect_roll_mean            388797 non-null  float64
 16  ambient_temp_effect_roll_std             388797 non-null  float64
 17  ambient_temp_effect_lag1                 388797 non-null  float64
 18  ambient_temp_effect_lag2                 388797 non-null  float64
 19  ambient_temp_effect_lag3                 388797 non-null  float64
 20  reactor_temp_roll_mean                   388797 non-null  float64
 21  reactor_temp_roll_std                    388797 non-null  float64
 22  reactor_temp_lag1                        388797 non-null  float64
 23  reactor_temp_lag2                        388797 non-null  float64
 24  reactor_temp_lag3                        388797 non-null  float64
 25  reactor_pressure_roll_mean               388797 non-null  float64
 26  reactor_pressure_roll_std                388797 non-null  float64
 27  reactor_pressure_lag1                    388797 non-null  float64
 28  reactor_pressure_lag2                    388797 non-null  float64
 29  reactor_pressure_lag3                    388797 non-null  float64
 30  feed_flow_rate_roll_mean                 388797 non-null  float64
 31  feed_flow_rate_roll_std                  388797 non-null  float64
 32  feed_flow_rate_lag1                      388797 non-null  float64
 33  feed_flow_rate_lag2                      388797 non-null  float64
 34  feed_flow_rate_lag3                      388797 non-null  float64
 35  coolant_flow_rate_roll_mean              388797 non-null  float64
 36  coolant_flow_rate_roll_std               388797 non-null  float64
 37  coolant_flow_rate_lag1                   388797 non-null  float64
 38  coolant_flow_rate_lag2                   388797 non-null  float64
 39  coolant_flow_rate_lag3                   388797 non-null  float64
 40  agitator_speed_rpm_roll_mean             388797 non-null  float64
 41  agitator_speed_rpm_roll_std              388797 non-null  float64
 42  agitator_speed_rpm_lag1                  388797 non-null  float64
 43  agitator_speed_rpm_lag2                  388797 non-null  float64
 44  agitator_speed_rpm_lag3                  388797 non-null  float64
 45  conversion_rate_roll_mean                388797 non-null  float64
 46  conversion_rate_roll_std                 388797 non-null  float64
 47  conversion_rate_lag1                     388797 non-null  float64
 48  conversion_rate_lag2                     388797 non-null  float64
 49  conversion_rate_lag3                     388797 non-null  float64
 50  selectivity_roll_mean                    388797 non-null  float64
 51  selectivity_roll_std                     388797 non-null  float64
 52  selectivity_lag1                         388797 non-null  float64
 53  selectivity_lag2                         388797 non-null  float64
 54  selectivity_lag3                         388797 non-null  float64
 55  yield_pct_roll_mean                      388797 non-null  float64
 56  yield_pct_roll_std                       388797 non-null  float64
 57  yield_pct_lag1                           388797 non-null  float64
 58  yield_pct_lag2                           388797 non-null  float64
 59  yield_pct_lag3                           388797 non-null  float64
 60  vibration_rms_roll_mean                  388797 non-null  float64
 61  vibration_rms_roll_std                   388797 non-null  float64
 62  vibration_rms_lag1                       388797 non-null  float64
 63  vibration_rms_lag2                       388797 non-null  float64
 64  vibration_rms_lag3                       388797 non-null  float64
```

```
 65  motor_current_roll_mean                  388797 non-null  float64
 66  motor_current_roll_std                   388797 non-null  float64
 67  motor_current_lag1                       388797 non-null  float64
 68  motor_current_lag2                       388797 non-null  float64
 69  motor_current_lag3                       388797 non-null  float64
 70  power_consumption_kw_roll_mean           388797 non-null  float64
 71  power_consumption_kw_roll_std            388797 non-null  float64
 72  power_consumption_kw_lag1                388797 non-null  float64
 73  power_consumption_kw_lag2                388797 non-null  float64
 74  power_consumption_kw_lag3                388797 non-null  float64
 75  efficiency_loss_pct_roll_mean            388797 non-null  float64
 76  efficiency_loss_pct_roll_std             388797 non-null  float64
 77  efficiency_loss_pct_lag1                 388797 non-null  float64
 78  efficiency_loss_pct_lag2                 388797 non-null  float64
 79  efficiency_loss_pct_lag3                 388797 non-null  float64
 80  time_elapsed_roll_mean                   388797 non-null  float64
 81  time_elapsed_roll_std                    388797 non-null  float64
 82  time_elapsed_lag1                        388797 non-null  float64
 83  time_elapsed_lag2                        388797 non-null  float64
 84  time_elapsed_lag3                        388797 non-null  float64
 85  ambient_temp_effect_x_coolant_flow_rate  388797 non-null  float64
 86  conversion_rate_x_feed_flow_rate         388797 non-null  float64
 87  conversion_rate_x_yield_pct              388797 non-null  float64
 88  selectivity_x_yield_pct                  388797 non-null  float64
 89  motor_current_x_vibration_rms            388797 non-null  float64
 90  ambient_temp_effect_x_time_elapsed       388797 non-null  float64
 91  coolant_flow_rate_x_time_elapsed         388797 non-null  float64
dtypes: float64(91), object(1)
memory usage: 272.9+ MB
```

Model (Baseline and Separation)

Baseline

Note: This model serves as a baseline template.

The tree parameters were constrained due to both fit and performance considerations. The following values were selected and can be fine-tuned in future work:

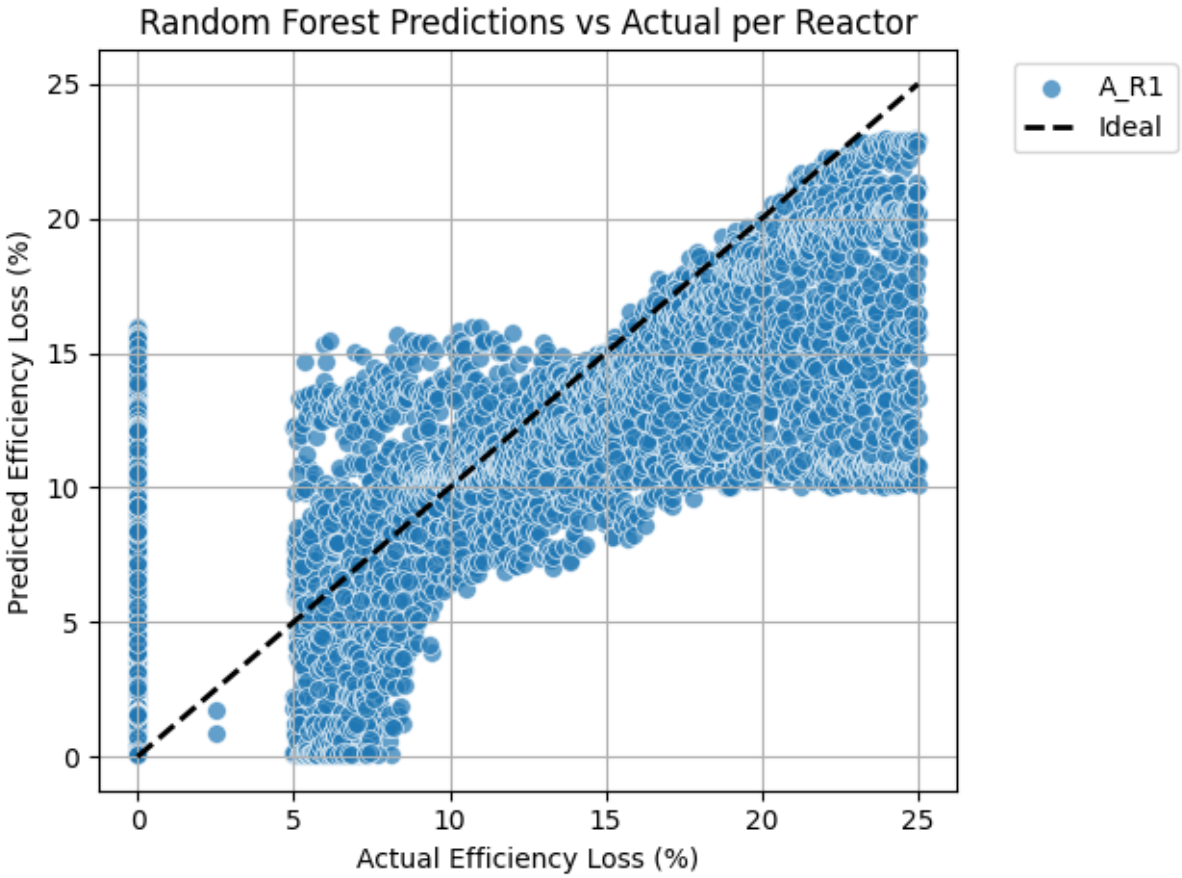
Parameter	Value	Reasoning
max_depth	5	Typical heuristic range is 10–15; significant overfitting was observed, so it was adjusted to 5 in practice
min_samples_leaf	5	Reduced from 10 to limit overfitting
min_samples_split	15	Reduced from higher values to improve performance
n_estimators	50	Reduced from 100 due to performance bottlenecks

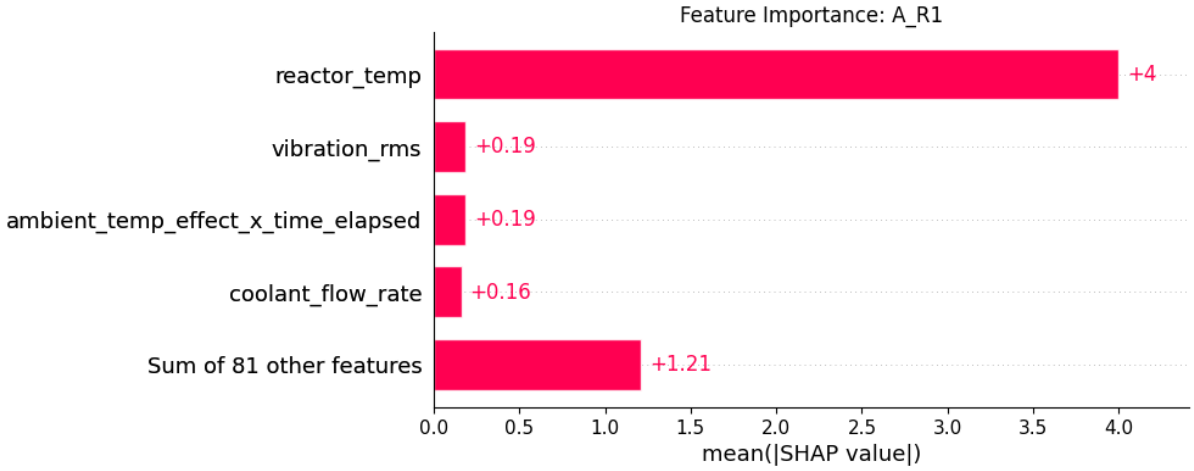
max_features	0.3	Increased from 0.1 to allow more tree complexity, since `n_estimators` was performance-limited
max_samples	0.25	Heuristic choice to increase tree diversity
n_jobs	-1	Uses all CPU cores for faster training
random_state	seed_1	Ensures reproducibility
oob_score	True	Enables out-of-bag validation

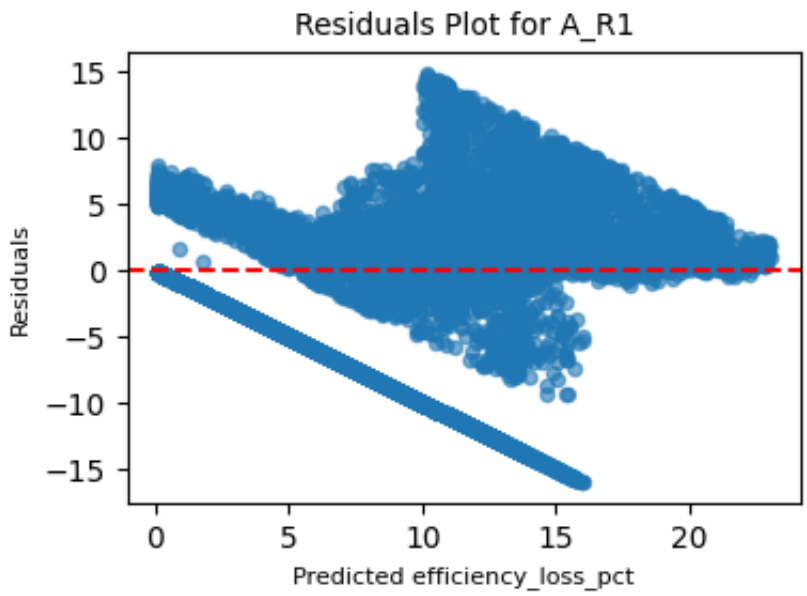
In [31]:
```
# Create Empty List for Results
results = []

# Loop Through Each Reactor
for test_reactor in reactor_names_A:

    # Split Train/Test by Reactor
    train_df = df_combined[df_combined['reactor_id'] != test_reactor].
    test_df  = df_combined[df_combined['reactor_id'] == test_reactor].

    # Remove reactor_id Column
    train_df = train_df.drop(columns=['reactor_id'])
    test_df  = test_df.drop(columns=['reactor_id'])

    # Identify Target-Related columns to Remove
    target_related_cols = [col for col in train_df.columns if 'efficie

    # Put Into Test and Train Variables
    X_train = train_df.drop(columns = target_related_cols)
    y_train = train_df['efficiency_loss_pct']

    X_test = test_df.drop(columns = target_related_cols)
    y_test = test_df['efficiency_loss_pct']

    # Set Up Random Forest
    reg = RandomForestRegressor(
        max_depth = 5,
        min_samples_leaf = 5,
        min_samples_split = 15,
        n_estimators = 50,
        max_features = 0.3,
        max_samples = 0.25,
        n_jobs = -1,
        random_state = seed_1,
        oob_score = True
    )

    # Fit the Model
    reg.fit(X_train, y_train)

    # Save
```

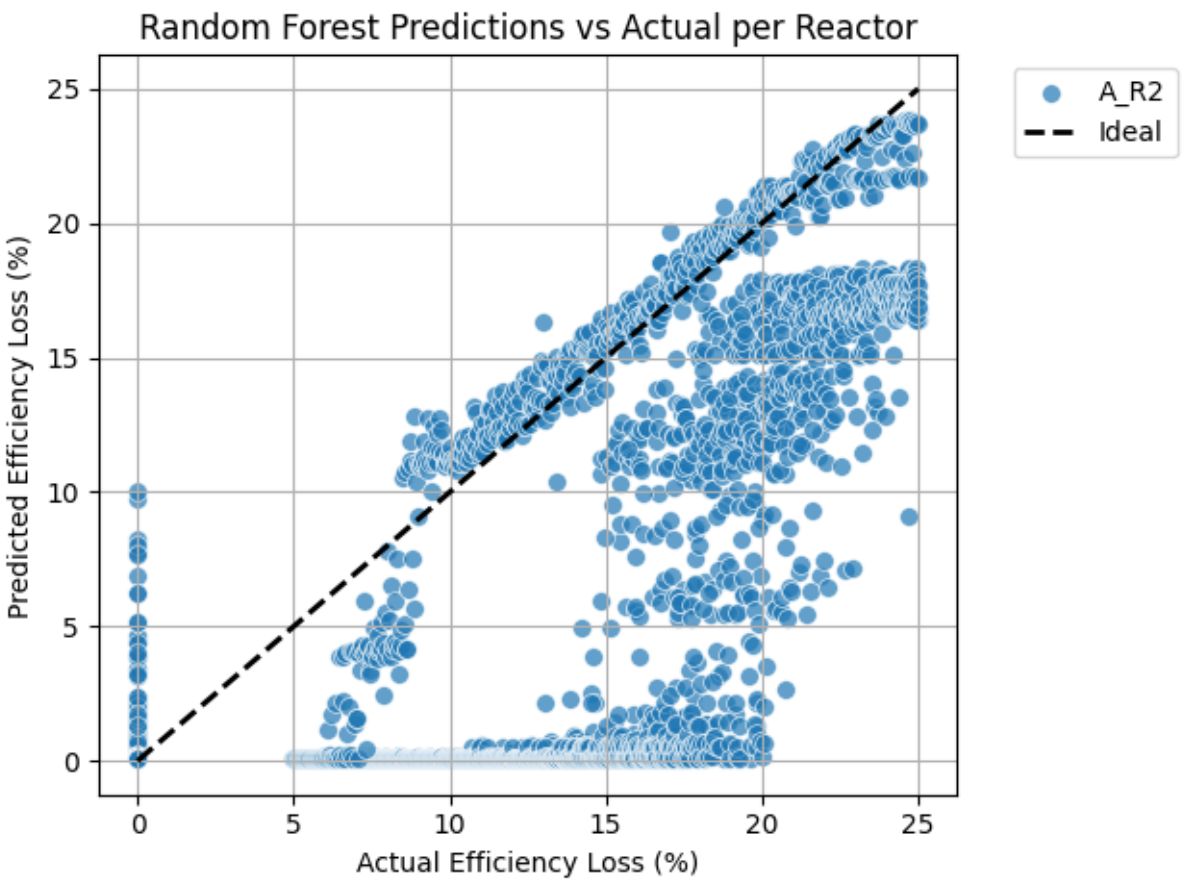
```
    model_filename = f"../model/model_base_{test_reactor}.joblib"
    joblib.dump(reg, model_filename)

    # Get Predictions
    y_train_pred = reg.predict(X_train)
    y_test_pred  = reg.predict(X_test)

    # Review Metrics
    results.append({
        'test_reactor': test_reactor,
        'train_mae': mean_absolute_error(y_train, y_train_pred),
        'train_mse': mean_squared_error(y_train, y_train_pred),
        'train_r2': r2_score(y_train, y_train_pred),
        'test_mae': mean_absolute_error(y_test, y_test_pred),
        'test_mse': mean_squared_error(y_test, y_test_pred),
        'test_r2': r2_score(y_test, y_test_pred),
        'oob_r2': reg.oob_score_
    })

    # Get Test Residuals
    residuals = y_test - y_test_pred

    # Prediction Scatter Plot
    sns.scatterplot(x=y_test, y=y_test_pred, label=test_reactor, s=50,

    # Plot the y=x line
    plt.plot([df_combined['efficiency_loss_pct'].min(), df_combined['e
            [df_combined['efficiency_loss_pct'].min(), df_combined['ef
            'k--', lw=2, label='Ideal')

    # Set Up Plot
    plt.xlabel('Actual Efficiency Loss (%)')
    plt.ylabel('Predicted Efficiency Loss (%)')
    plt.title('Random Forest Predictions vs Actual per Reactor')
    plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
    plt.grid(True)
    plt.tight_layout()
    plt.show()

    # Get SHAP
    explainer = shap.TreeExplainer(reg)
    shap_ex = explainer(X_test, check_additivity=False)

    # Plot Feature Importance
    plt.title(f"Feature Importance: {test_reactor}")
    shap.plots.bar(shap_ex, max_display = 5)
    plt.show()

    # Residuals Plot
    plt.figure(figsize=(4,3))
    plt.scatter(y_test_pred, residuals, s=20, alpha=0.6)
    plt.axhline(0, color='red', linestyle='--')
```

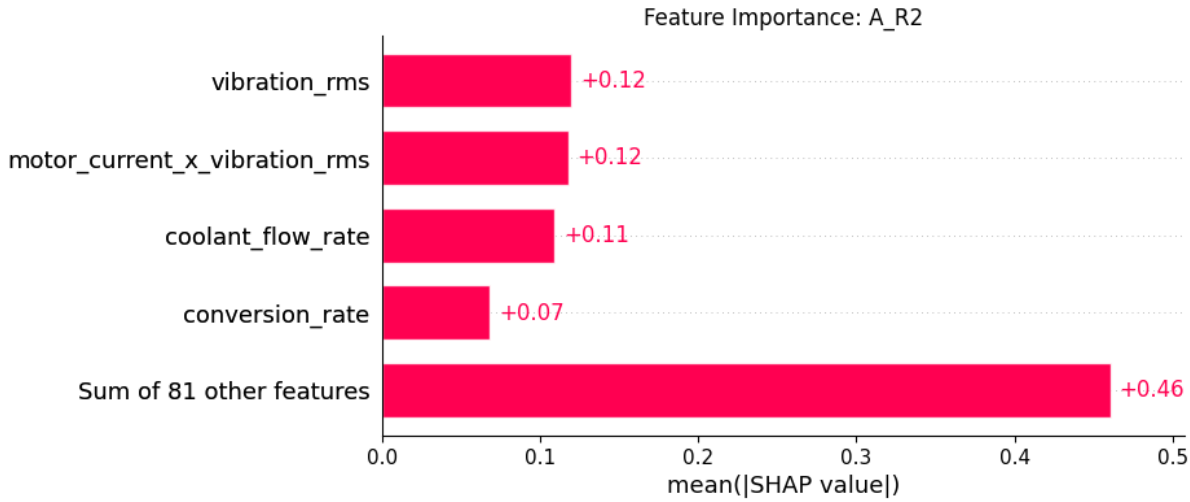
```
    plt.xlabel("Predicted efficiency_loss_pct", fontsize=8)
    plt.ylabel("Residuals", fontsize=8)
    plt.title(f"Residuals Plot for {test_reactor}", fontsize=10)
    plt.tight_layout()
    plt.show()

# Convert to DataFrame
results_df = pd.DataFrame(results)

# Display R2 Results
results_df = results_df.sort_values('test_r2', ascending=False)
print(results_df)
```

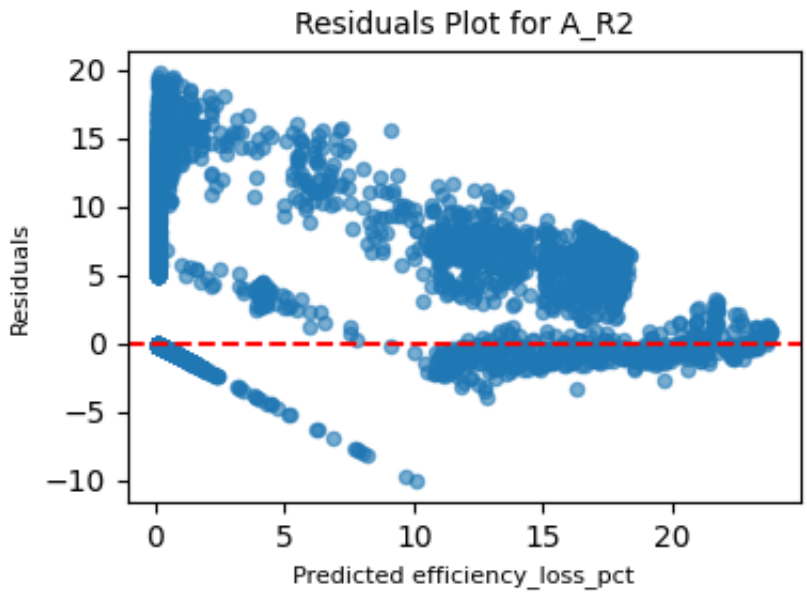


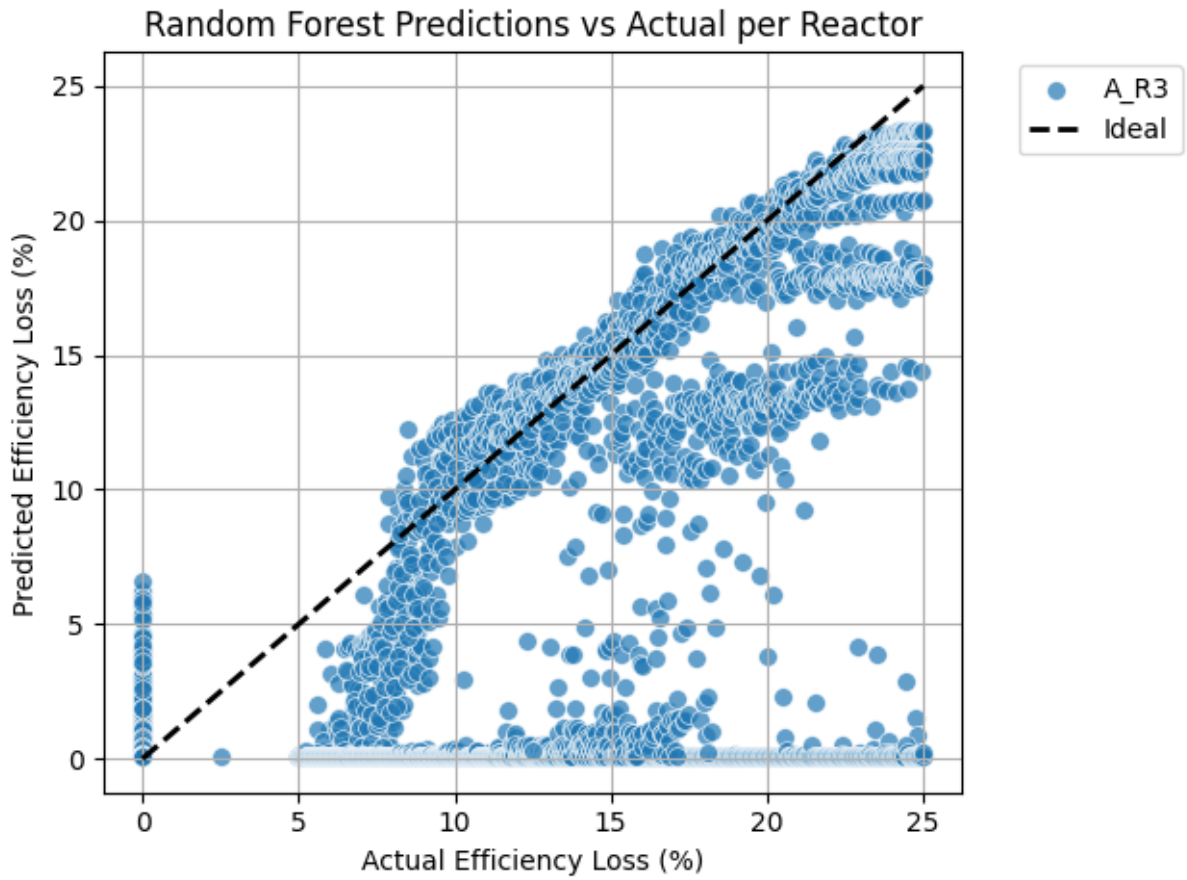


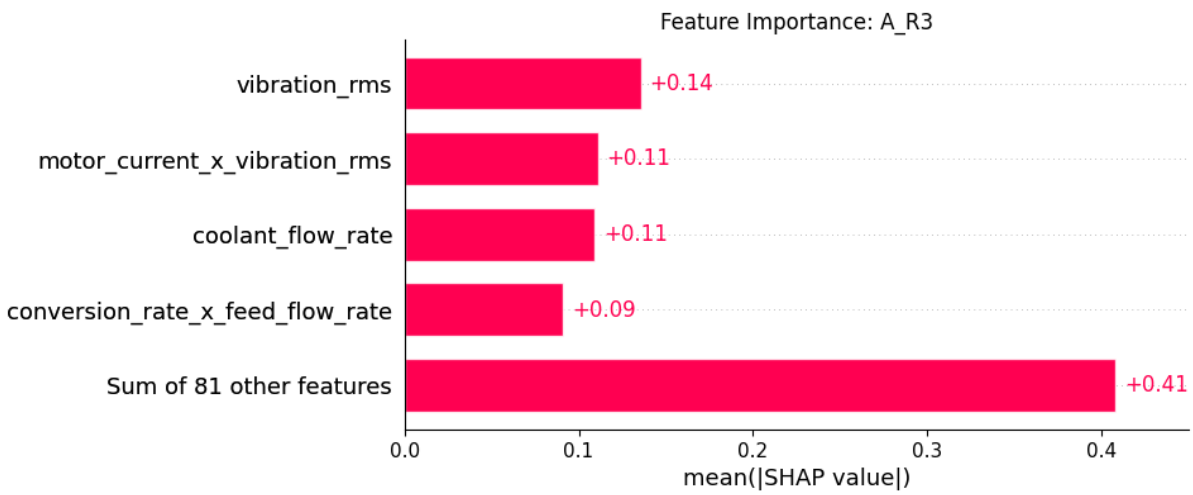

Residuals Plot for A_R1
Residuals
Predicted efficiency_loss_pct
15
10
5
0
−5
−10
−15
0
5
10
15
20

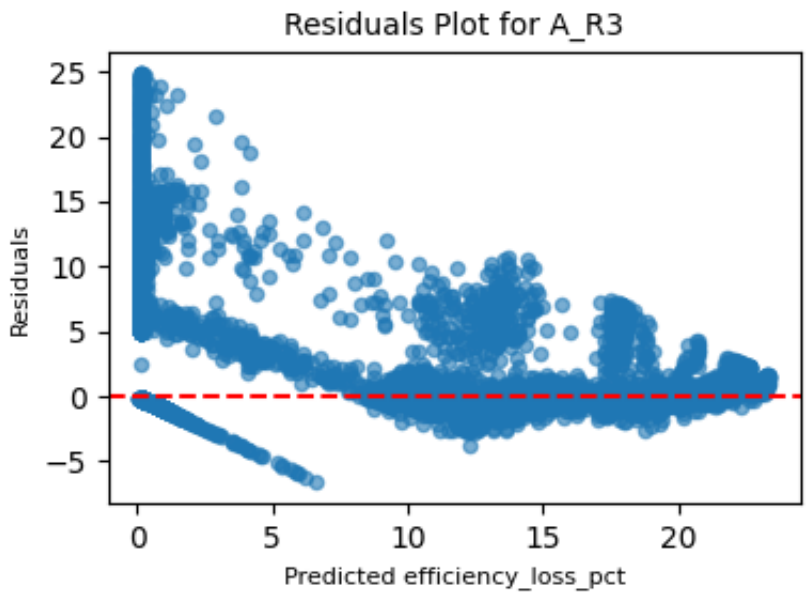

Random Forest Predictions vs Actual per Reactor
Predicted Efficiency Loss (%)
Actual Efficiency Loss (%)
A_R2
Ideal
25
20
15
10
5
0
0
5
10
15
20
25


Feature Importance: A_R2
vibration_rms
motor_current_x_vibration_rms
coolant_flow_rate
conversion_rate
Sum of 81 other features
+0.12
+0.12
+0.11
+0.07
+0.46
0.0
0.1
0.2
0.3
0.4
0.5
mean(|SHAP value|)


Residuals Plot for A_R2
Residuals
20
15
10
5
0
−5
−10
0
5
10
15
20
Predicted efficiency_loss_pct


Random Forest Predictions vs Actual per Reactor
A_R3
Ideal
Predicted Efficiency Loss (%)
Actual Efficiency Loss (%)
0
5
10
15
20
25


Feature Importance: A_R3
vibration_rms
+0.14
motor_current_x_vibration_rms
+0.11
coolant_flow_rate
+0.11
conversion_rate_x_feed_flow_rate
+0.09
Sum of 81 other features
+0.41
0.0
0.1
0.2
0.3
0.4
mean(|SHAP value|)




```
  test_reactor  train_mae  train_mse  train_r2  test_mae   test_mse   t
est_r2  \
2         A_R3   0.223353   0.957158  0.893608  0.332937   2.788773  0.
691361
1         A_R2   0.274184   1.498072  0.858109  0.321215   2.008123  0.
657149
0         A_R1   0.241149   1.106987  0.851528  3.475932  27.171019 -1.
252593

     oob_r2
2  0.892351
1  0.856977
0  0.849955
```

Note: All models score highly when predicting unseen sensor data, as shown by OOB performance. However, each model appears to learn reactor-specific rules that do not transfer easily across units. Feature importance suggests that `A_R1` is highly susceptible to changes in `reactor_temp`, while other reactors show signs of higher complexity and potential overfitting.

Based on R2 values, `A_R1` is an outlier: its predictive performance drops more when applied to other reactors. `A_R3` performs best, capturing the physical trend and transferring relatively better to other units. `A_R2` is intermediate, with moderate transferability. Residual patterns indicate that the model does not fully capture reactor-specific behavior and may be missing a linear component in the system.

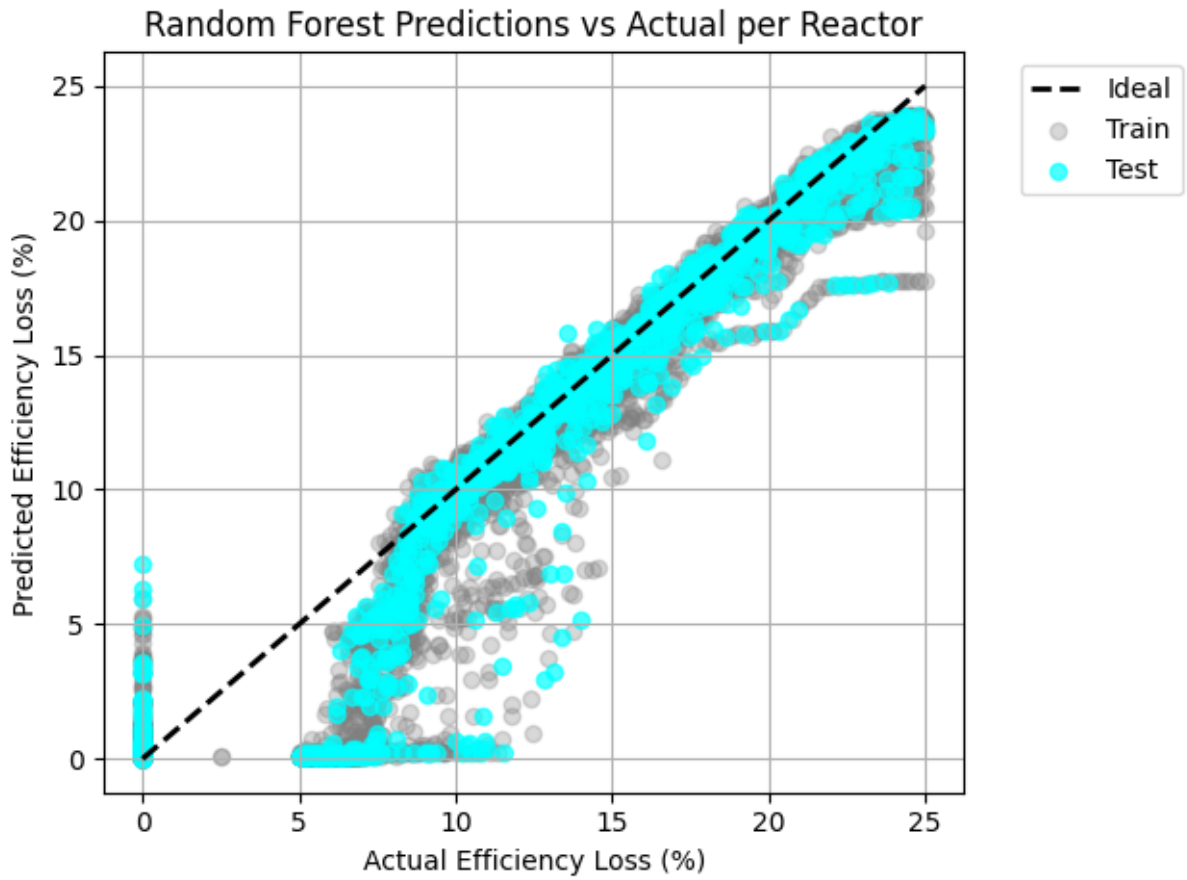
MSE and MAE show a similar pattern for `A_R1`: MAE increases from train to test, and MSE rises substantially, indicating larger outlier errors in test predictions. For `A_R2` and `A_R3`, this increase is less severe, suggesting better capture of feature-target relationships.

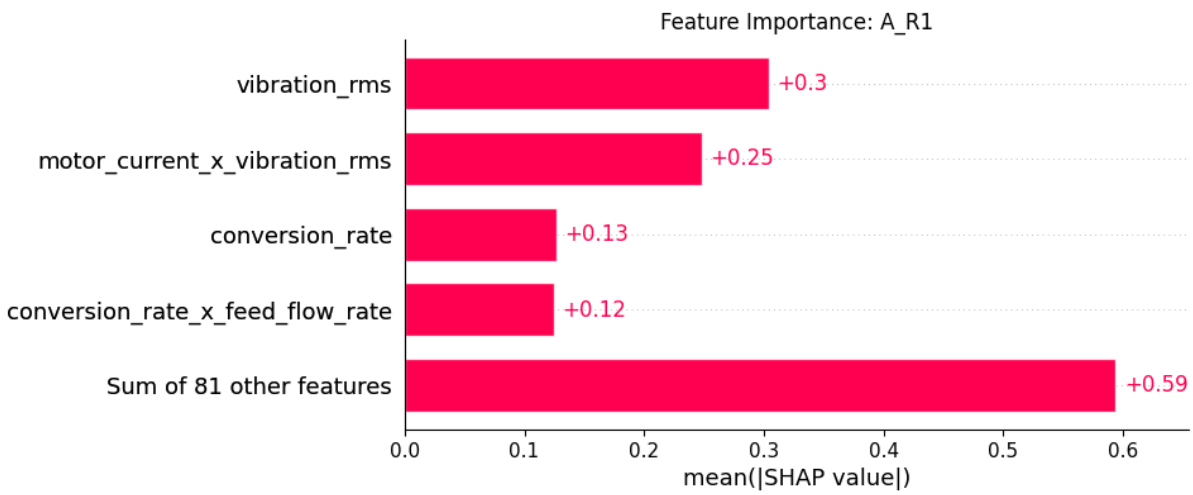
Independent models for each reactor will be considered next.

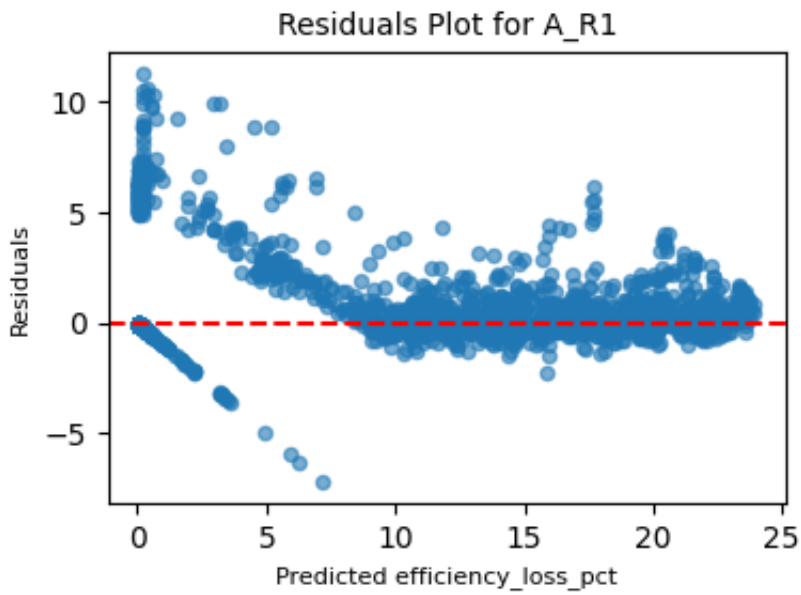
Model (Separate)

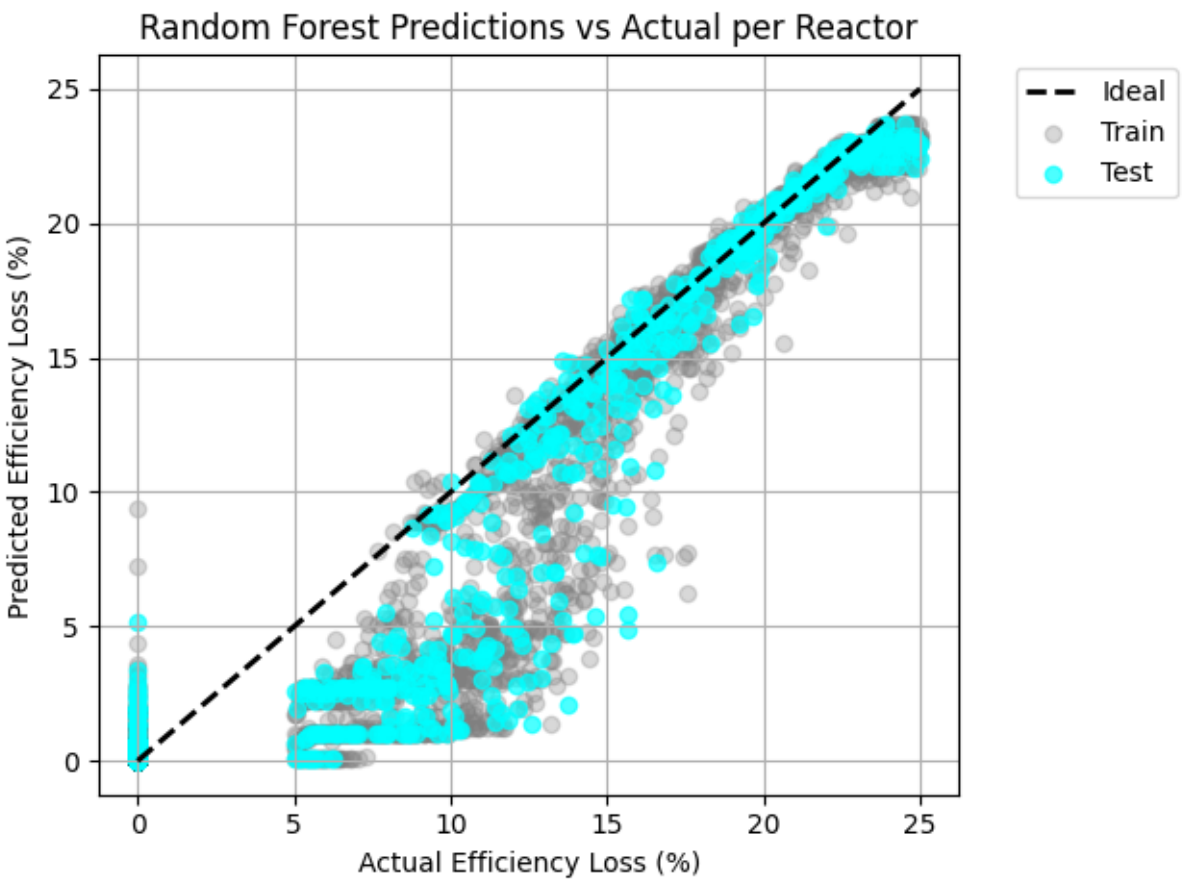
In [32]:
```
# Create Empty List and Dictionary for Results
results_sep = []
models = {}

# Loop Through Each Reactor
for reactor in reactor_names_A:

    # Get Based on Reactor ID
    reactor_df = df_combined[df_combined['reactor_id'] == reactor].cop

    # Remove Target Columns
    target_related_cols = [col for col in train_df.columns if 'efficie

    X = reactor_df.drop(columns=['reactor_id'] + target_related_cols)
    y = reactor_df['efficiency_loss_pct']

    # 80/20 Train-Test Split
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size = 0.20, random_state = seed_1
    )

    # Set Up Random Forest
    reg = RandomForestRegressor(
        max_depth = 5,
        min_samples_leaf = 5,
        min_samples_split = 15,
        n_estimators = 50,
        max_features = 0.3,
        max_samples = 0.25,
        n_jobs = -1,
        random_state = seed_1,
        oob_score = True
    )

    # Evaluate Model on Training Data
    reg.fit(X_train, y_train)
    models[reactor] = reg

    # Save model
    joblib.dump(reg, f"../model/model_sep_{reactor}.joblib")

    # Get Predictions for Test and Train
    y_train_pred = reg.predict(X_train)
    y_test_pred  = reg.predict(X_test)

    # Append Results to Output
    results_sep.append({
```

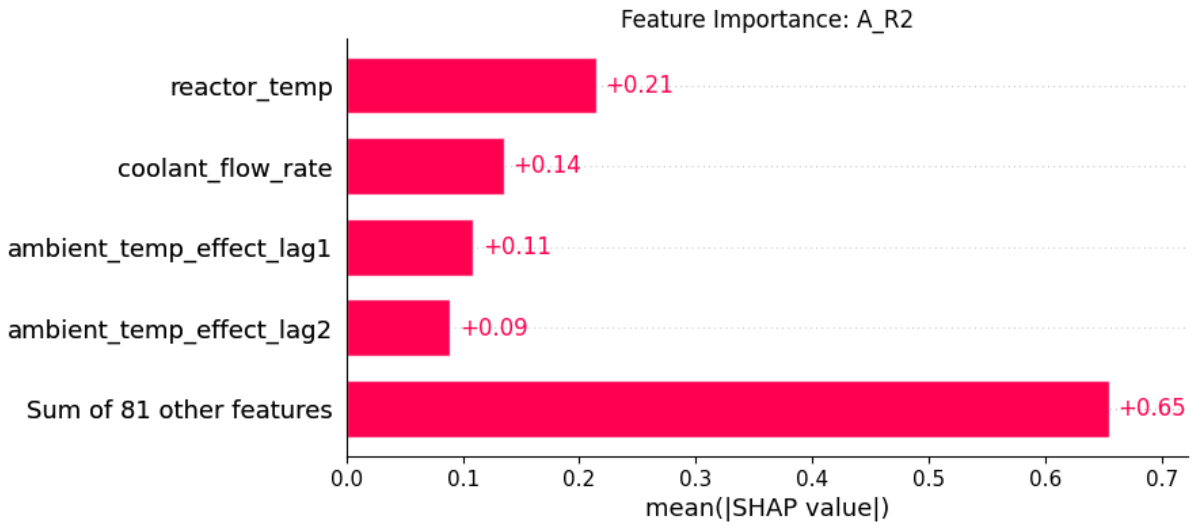
```
        'reactor': reactor,
        'train_r2': r2_score(y_train, y_train_pred),
        'test_r2': r2_score(y_test, y_test_pred),
        'oob_r2': reg.oob_score_,
        'test_mae': mean_absolute_error(y_test, y_test_pred)
    })

    # Get Residuals
    residuals = y_test - y_test_pred

    # Plot the y=x line
    plt.plot([df_combined['efficiency_loss_pct'].min(), df_combined['e
            [df_combined['efficiency_loss_pct'].min(), df_combined['ef
            'k--', lw=2, label='Ideal')
    plt.scatter(y_train, y_train_pred, alpha=0.3, label='Train', color
    plt.scatter(y_test, y_test_pred, alpha=0.7, label='Test', color='c

    # Set Up Plot
    plt.xlabel('Actual Efficiency Loss (%)')
    plt.ylabel('Predicted Efficiency Loss (%)')
    plt.title('Random Forest Predictions vs Actual per Reactor')
    plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
    plt.grid(True)
    plt.tight_layout()
    plt.show()

    # Get SHAP
    explainer = shap.TreeExplainer(reg)
    shap_ex = explainer(X_test, check_additivity=False)

    # Plot Feature Importance
    plt.title(f"Feature Importance: {reactor}")
    shap.plots.bar(shap_ex, max_display = 5)
    plt.show()

    # Residuals Plot
    plt.figure(figsize=(4,3))
    plt.scatter(y_test_pred, residuals, s=20, alpha=0.6)
    plt.axhline(0, color='red', linestyle='--')
    plt.xlabel("Predicted efficiency_loss_pct", fontsize=8)
    plt.ylabel("Residuals", fontsize=8)
    plt.title(f"Residuals Plot for {reactor}", fontsize=10)
    plt.tight_layout()
    plt.show()

# Final Results
df_sep = pd.DataFrame(results_sep)
print(df_sep)
```

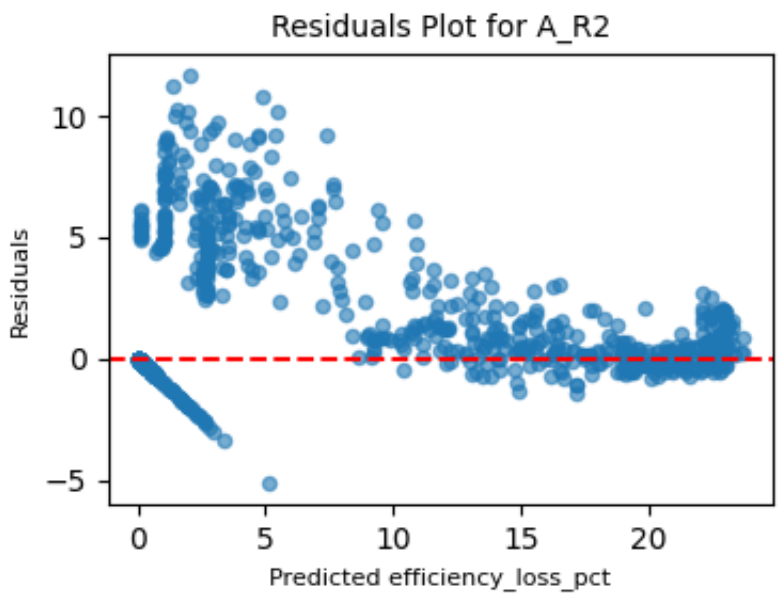

Random Forest Predictions vs Actual per Reactor
Ideal
Train
Test
Predicted Efficiency Loss (%)
Actual Efficiency Loss (%)
0
5
10
15
20
25

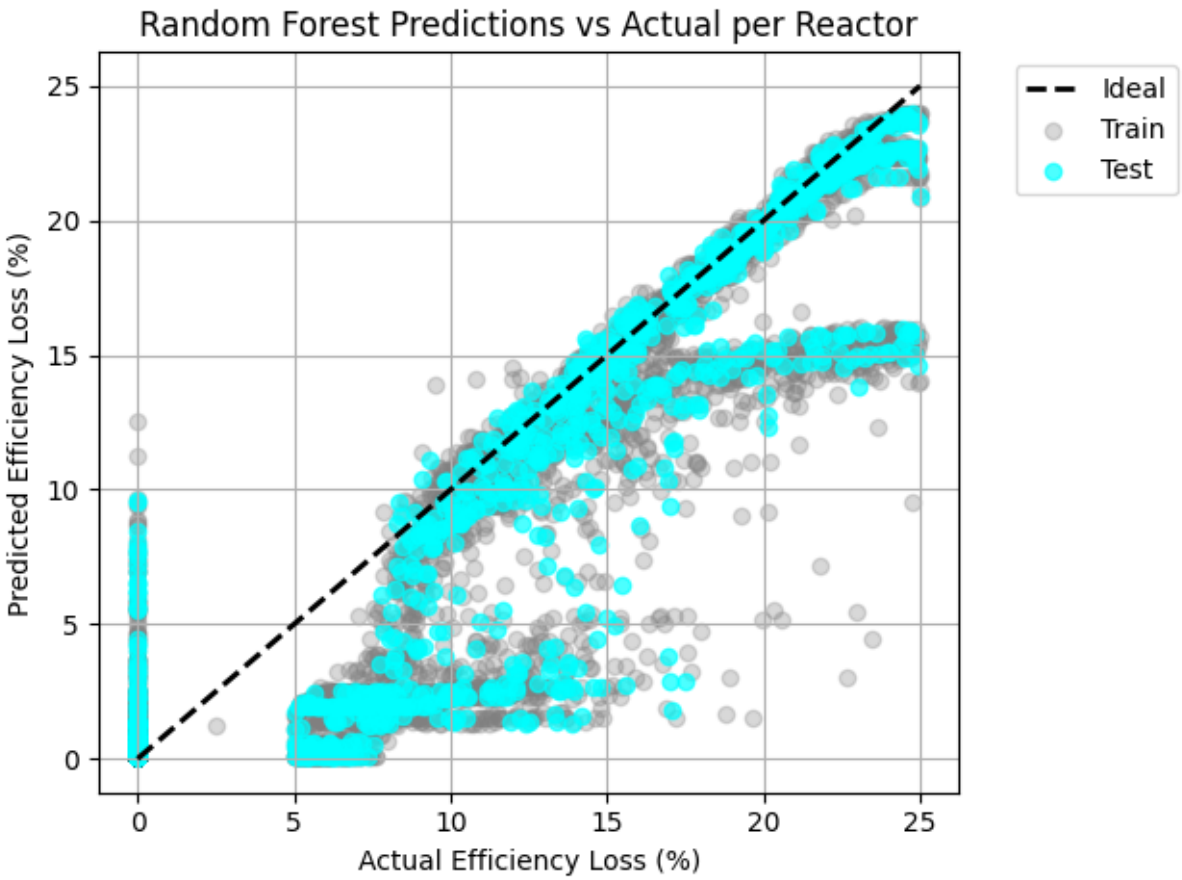

Feature Importance: A_R1
vibration_rms
motor_current_x_vibration_rms
conversion_rate
conversion_rate_x_feed_flow_rate
Sum of 81 other features
+0.3
+0.25
+0.13
+0.12
+0.59
0.0
0.1
0.2
0.3
0.4
0.5
0.6
mean(|SHAP value|)

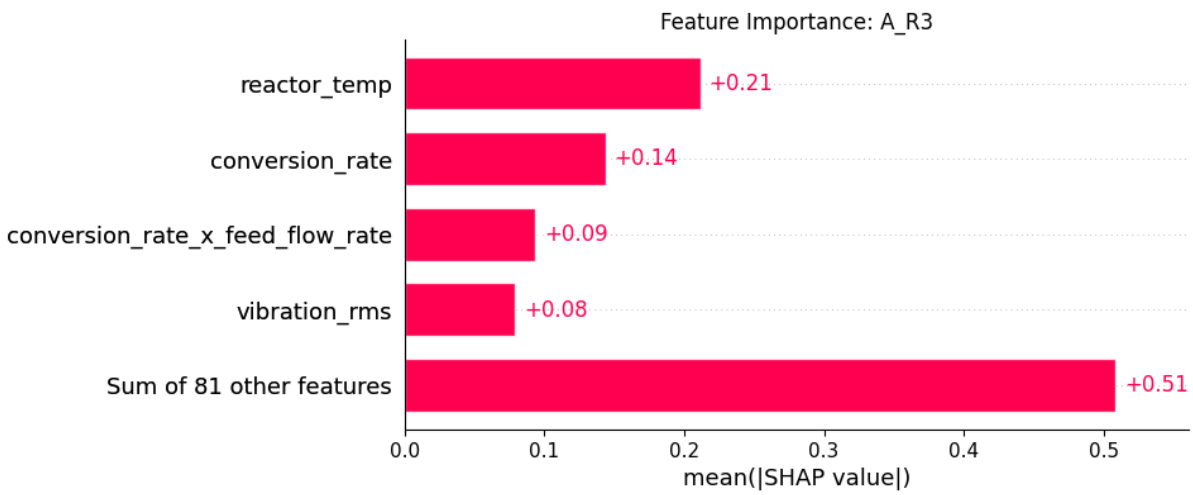

Residuals Plot for A_R1
Residuals
Predicted efficiency_loss_pct
10
5
0
−5
0
5
10
15
20
25

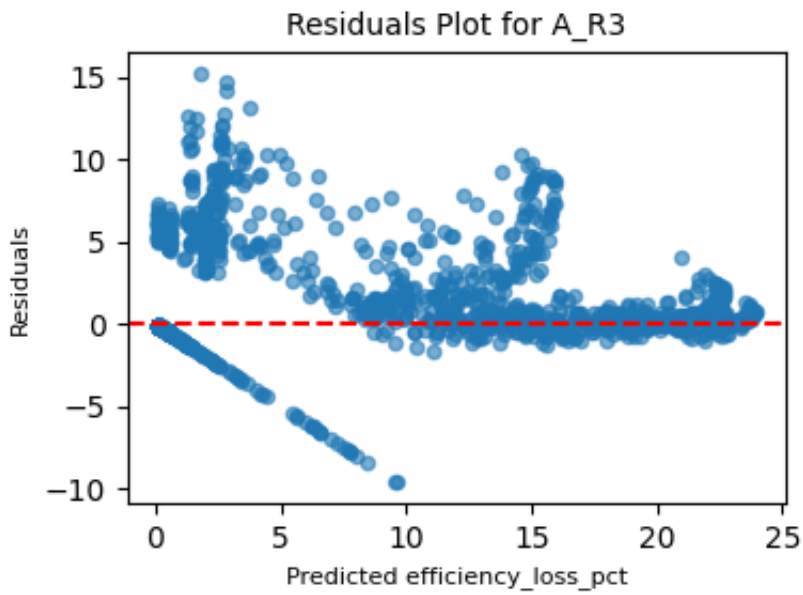

Random Forest Predictions vs Actual per Reactor
Predicted Efficiency Loss (%)
Actual Efficiency Loss (%)
Ideal
Train
Test
25
20
15
10
5
0
0
5
10
15
20
25


Feature Importance: A_R2
reactor_temp
coolant_flow_rate
ambient_temp_effect_lag1
ambient_temp_effect_lag2
Sum of 81 other features
+0.21
+0.14
+0.11
+0.09
+0.65
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
mean(|SHAP value|)


Residuals Plot for A_R2
Residuals
10
5
0
−5
0
5
10
15
20
Predicted efficiency_loss_pct


Random Forest Predictions vs Actual per Reactor
Ideal
Train
Test
Predicted Efficiency Loss (%)
Actual Efficiency Loss (%)
0
5
10
15
20
25


Feature Importance: A_R3
reactor_temp
+0.21
conversion_rate
+0.14
conversion_rate_x_feed_flow_rate
+0.09
vibration_rms
+0.08
Sum of 81 other features
+0.51
0.0
0.1
0.2
0.3
0.4
0.5
mean(|SHAP value|)




```
  reactor  train_r2   test_r2    oob_r2  test_mae
0    A_R1  0.971086  0.970646  0.970307  0.145991
1    A_R2  0.940208  0.936865  0.938914  0.121296
2    A_R3  0.920864  0.923703  0.918651  0.204335
```

Note: The training and test R2 values are close, which suggests limited overfitting. OOB scores are high, indicating that the modeling approach works well within the same reactor type. Feature importance suggests that `efficiency_loss_pct` is primarily influenced by chemical imbalance or motor issues for `A_R1`, temperature sensitivity for `A_R2`, and a combination of factors for `A_R3`.

Residual plots show better performance at higher target values than at lower ones. This indicates that the model struggles more with predicting small `efficiency_loss_pct` values.

Conclusion

Initially, a Random Forest model was used to focus on Regime A in a leave-one-reactor-out simulation. With conflicting R2 values and predictions, and with `A_R1` acting as an outlier that is highly susceptible to `reactor_temp`, the model did not generalize well across reactors.

After shifting to independent reactor models, cross-reactor generalization was no longer required, and the new models captured reactor-specific vulnerabilities more effectively when predicting `efficiency_loss_pct`. This improvement was reflected in R2, MAE, and MSE results.

A key limitation is the use of 80+ features, many of which dominate feature

importance and increase computational cost. Reducing feature count and focusing on the most informative drivers would likely improve both model quality and runtime.

There is also bias at lower efficiency-loss levels, where the model struggles to reason through small values. Applying a log-linear approach for lower-range targets may help the model capture structure within that noise region.

Potential Work

- Dimensionality reduction (PCA) to simplify computation and reduce runtime
- Hyperparameter tuning to improve accuracy and efficiency
- Downsampling to reduce noise and improve performance
- Per-unit calibration (standardization), especially for other model families
- Evaluating additional tree-based models beyond `RandomForestRegressor`
- Examining abrupt sensor changes to detect potential machine defects
- Including additional columns, such as setpoints
- Applying physical models
  - Arrhenius equation
  - Conservation of energy
  - Mass balances
  - Ohm's law
- Exploring temperature seasonality
- Additional improvement areas
  - Regime B analysis
  - Value anomaly detection
  - Fault classification